Executive Summary

This document summarizes a structural tension identified in the standard formulation of the Yang–Mills mass gap problem as posed by the Clay Mathematics Institute. The standard reading of the problem assumes: (1) a unique Poincaré-invariant vacuum state, and (2) a mass gap intrinsic to the four-dimensional Yang–Mills theory itself. However, widely accepted observations concerning vacuum energy indicate that vacuum quantities are accessed through relative measurements rather than as absolute internal values. If mass gaps are permitted to arise from relative or interlayer differences, the assumption that the gap must be intrinsic to a single four-dimensional vacuum becomes structurally restrictive. Taken together, these elements produce a logical tension in the standard axiomatic framing of the problem. This summary does not propose a solution to the mass gap problem, nor does it introduce a new physical model. Its sole purpose is to clarify the assumptions implicit in the current formulation and to identify where those assumptions may conflict with minimal physical observations.